Dynamic-Hand-Gesture Authentication Dataset and Benchmark

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***Abstract*— In recent years, biometrics have received considerable attention for its reliability and usability. Dynamic-hand-gesture is one of the representative biometric modalities, with advantages of safety and template-replaceability, has huge potential value. However, due to the lack of large-scale dataset and comprehensive evaluation methods, few researches are intended to study the dynamic-hand-gesture authentication method. In this article, we introduce a new dataset SCUT-DHGA, which is the first large-scale Dynamic-Hand-Gestures-Authentication dataset. SCUT-DHGA contains 29,160 dynamic-hand-gesture video sequences and more than 1.86 million frames for both color and depth modalities acquired from 193 volunteers. Six kinds of dynamic-hand-gestures are carefully designed for researching two types of authentication tasks: gesture-predefined authentication and gesture-free authentication. To investigate the hypothesis that users' gestures would be variant after time-span, which will degrade the performance of a dynamic-hand-gesture authentication system, two separate sessions' data were acquired from 50 volunteers with an average interval of one week. Beside the SCUT-DHGA dataset, we also benchmark this dataset with our proposed DHGA-net. By releasing such a large-scale dataset and benchmark, we expect dynamic-hand-gesture authentication methods to gain further improvement and generalization.**

***Index Terms*— Dynamic-hand-gesture, authentication, dataset, two-sessions, benchmark.**

I. Introduction

HAND-GESTURE can be used to express instruction, emotion, and also can be used as a kind of biometric modalities. Many efforts have been dedicated to hand-gesture recognition, which try to classify different hand-gestures. The ultimate aim of hand-gesture recognition is to understand the semantic of gestures and using gestures for Human Computer Interaction (HCI). As a new emerging biometric modality, hand-gesture is template renewable, and has great research value and wide application potentiality. In this work, we focus on hand-gesture authentication, with the purpose of distinguishing different people by using hand-gesture data. **It is worth to note that, hand-gesture authentication and hand-gesture recognition are essential different tasks. For convenience, "hand-gesture recognition" will be referred to as "hand-gesture classification" in this article**.

Biometrics are a technical term for body measurements and calculations, they refer to metrics related to human characteristics. Since biological features are the distinctive, measurable characteristics used to label and describe individuals, they are commonly used in user identification and access control [1]. Compare to traditional access control methods such as knowledge-based methods (*e.g.*, passwords and answer security questions) and token-based methods (*e.g.*, keys and ID cards), biometrics cannot be forgotten since they rely solely on user-specific traits, therefore they are more reliable and practical [2].

Generally, biometric identifiers are often categorized as physiological versus behavioral characteristics [3]. Physiological characteristics are related to the physiological uniqueness of individuals, such as fingerprint, palm print, hand geometry, palm veins, face, DNA, iris, retina and odour/scent. Behavioral characteristics are related to the pattern of personal behavior, such as typing rhythm, gait, and voice [4].

Although the potential authentication performance of dynamic-hand-gesture has not been fully developed yet, we consider that it contains both the advantages of physiological traits and behavioral traits. Firstly, hand shape and skin texture of palm can be observed in each dynamic-hand-gesture video, representing the user's physiological characteristics. Secondly, finger movement habit is unique from user to user, which can be treated as behavioral characteristics. Thirdly, different from other physiological characteristics, dynamic-hand-gesture password is easy to change when compromised. Which is to say, when a hand-gesture for authentication has a risk to be stolen or not being safe, user could simply employ another hand-gesture, without needing to replace the entire system like other authentication system based on physiological traits.

To explain the hand-gesture-authentication more clearly, the upper part of Fig.1 shows the flow chart of a dynamic-hand-gesture authentication system. Given a **query** hand-gesture and **claimed** identity, a designed feature extractor is used to obtain the query feature. If the query feature is close in distance to a feature of the same identity in registered features database, it should be accepted as that user, otherwise it should be rejected.

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Data is available on-line at https://github.com/SCUT-BIP-Lab/SCUT-DHGA
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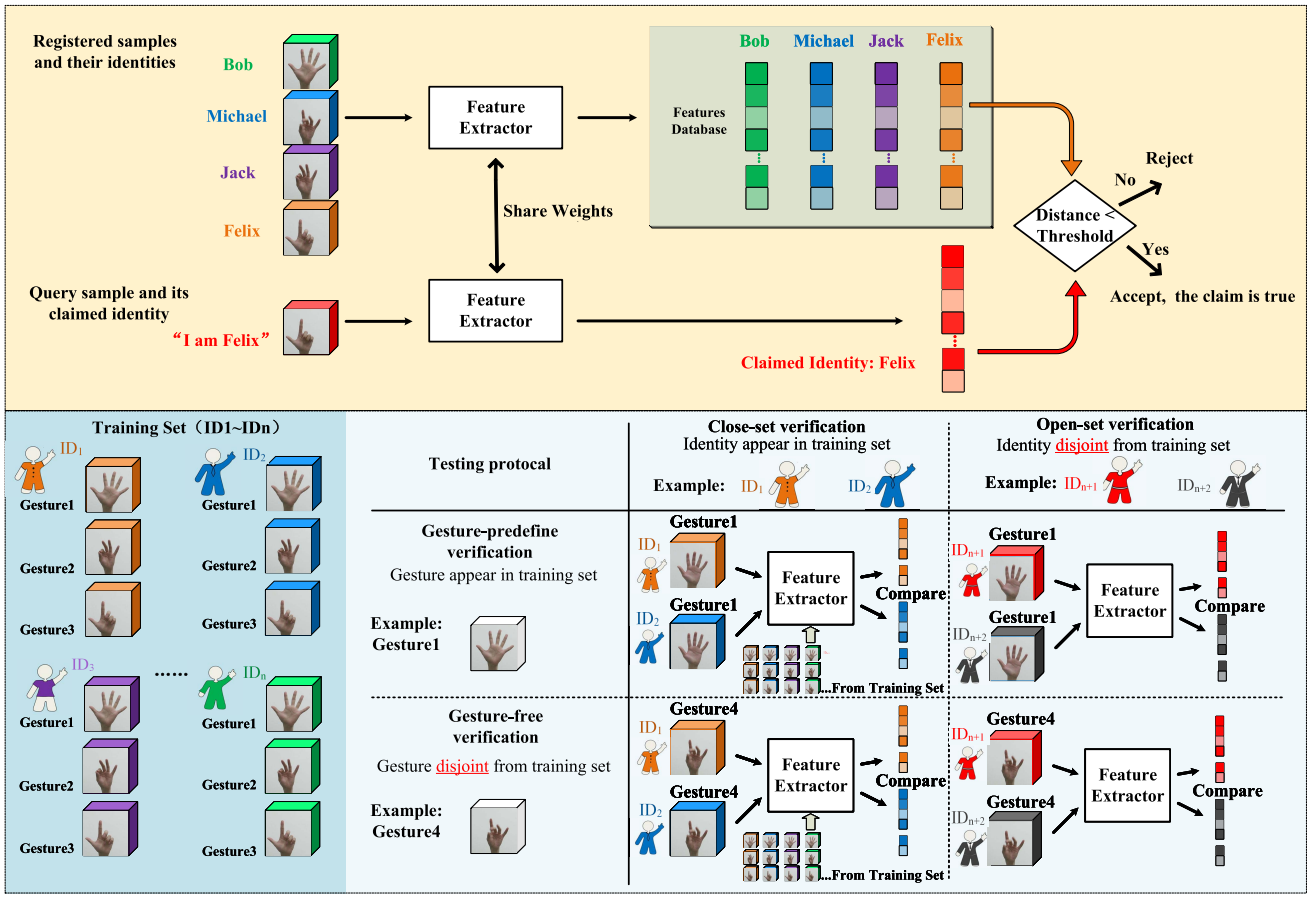


Fig. 1. Definition of gesture authentication and different testing protocols.

In terms of testing protocol, dynamic-hand-gesture authentication can be evaluated under **closed-set** or **open-set** settings, as illustrated in the second row of Fig.1. For closed-set authentication, the testing identities appear in training set, for open-set authentication, the testing identities are disjoint from the training set, which is more challenging yet close to practice. Besides, there are two kinds of dynamic-hand-gesture authentication: **gesture-predifined** authentication and **gesture-free** authentication.

For the former, all gesture types used for testing are seen by the feature extractor, therefore appear in the training set.

As for the latter, the gesture types used for testing are unseen by the feature extractor, they are unnecessary to appear in training set, and thus users can freely design their own gestures for authentication. Among these two ways, gesture-free authentication is more realistic and challenging. In this article, we test the performance of both the gesture-predifined and gesture-free authentication under open-set protocol.

There are mainly two types of framework for hand-gesture authentication, trajectory-based methods and vision-based methods. **Trajectory-based methods** use depth images or specific devices such as leap motion or RealSense to obtain key-points of hand from each video frame and tracking these key-points from frame to frame to obtain trajectories. These trajectories can be regarded as behavioral traits. Various algorithms can used to extract features from these trajectories.

Vision-based methods extract users' distinguishing features directly from image or video. Intuitively, vision-based methods take advantage of more user-specific characteristics than trajectory-based methods, because they explicitly take in more information (such as hand shape, skin texture, and fingers movement, etc.). We mainly focus on vision-based dynamic-hand-gesture authentication in this article.

There exist many large-scale hand-gesture datasets, but most of them are designed for hand-gesture classification. These datasets contain large amount of hand-gesture samples but from few subjects, like DHG-14/28 dataset [5], Sheffield Kinect Gesture (SKIG) Dataset [6], EgoGesture dataset [7], [8]. They are not appropriate for evaluation of gesture authentication because of the limited numbers of subject classes and cross-session samples in these datasets. To the best of our knowledge, the datasets specifically designed for hand-gesture authentication are scarce. Some of them contain only trajectories data [9], [10], which are not able to develop vision-based authentication methods. Some of them only contain depth images [11], not able to develop a hand-gesture authentication system working on portable devices without a depth sensor. Moreover, all of the existing hand-gesture datasets designed for authentication contain limited number of samples (no more than 1,700).

In this article, we introduce the up to date largest dynamic-hand-gesture authentication dataset SCUT-DHGA,

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which will be publicly available soon. Compared to other existing hand-gesture authentication datasets, SCUT-DHGA has following advantages. Firstly, SCUT-DHGA is large-scale, including 29,160 dynamic-hand-gesture video sequences for both RGB modality and depth modality, with more than 1.86 million RGB images and depth images which collected from 193 subjects of different ages and genders. Secondly, two sessions data were acquired to test the performance of using data with time-span. Thirdly, we propose a new DHGA-net to benchmark SCUT-DHGA. The equal error rate (EER) in three scenarios is tested to evaluate the authentication performance: 1) hand-gesture authentication in single session; 2) hand-gesture authentication in cross session; 3) hand-gesture authentication using multiple modality fusion. We believe our dataset can be used as a benchmark and help the vision-based dynamic-hand-gesture authentication to step forward.

II. Related Work

A. Hand-Gesture Datasets

Currently, there exist many established benchmark datasets of dynamic hand-gesture classification (not for authentication that the purpose is to distinguish identities, but for classifying different hand gesture types). The DHG-14/28 Dataset [5] is a commonly used benchmark gesture dataset with clean backgrounds, which contains 14 gestures performed in two ways: using one finger and the whole hand. Each gesture is performed 5 times by 20 participants, resulting in 2,800 sequences. Each frame of sequences contains a depth image and the coordinates of the hand skeleton.

Sheffield Kinect Gesture (SKIG) Dataset [6] contains 1,080 RGB sequences and 1,080 depth sequences from 6 subjects collected with a Kinect sensor. It has 10 gesture categories with 3 hand postures, 3 backgrounds, and 2 illumination conditions. EgoGesture Dataset [7], [8] is a large multimodal hand-gesture dataset collected for the task of egocentric gesture classification with wearable devices, it contains 24,161 hand-gesture clips of 83 classes of gestures from 50 distinct subjects. NVGestures Dataset [12] and VIVA Hand-Gestures Dataset [13] are two multimodal hand-gesture datasets with in-car scenes for HCI. NVGestures Dataset [12] is captured with multiple sensors and from multiple viewpoints. It contains 25 classes, 1,532 dynamic hand-gestures recorded from 20 subjects inside a car simulator with artificial lighting conditions. VIVA Hand-Gesture Dataset [13] contains 885 visible RGB and depth video sequences of 19 hand-gesture classes from 8 subjects. It is specifically designed with difficult settings of cluttered background, volatile illumination and frequent occlusion in real-world driving settings. Datasets mentioned above are all designed for gesture classification. These hand-gesture datasets always focus on distinguish different hand-gestures but ignore whose hand it is. Therefore, these datasets contain many types of gestures but from few subjects in general. But in authentication domain, gesture datasets should contain enough subjects instead of many gesture types. We think the datasets mentioned above are not suitable for evaluating hand-gesture authentication methods.

To the best of our knowledge, there are few established benchmark datasets in hand-gesture authentication domain. Some work([14], [15]) collected **static hand images** for authentication. In [14], the authors collected the right hand images of their colleagues using a very low cost scanner device to form a hand dataset: JUET contact database, which contains 50 subjects and each subject has 10 templates. For acquiring the static hand-gesture data, [15] gathered 4 student volunteers and each to generate four sets of hand-gestures for the 26 letters according to the standard American Sign Language. Some work collected **dynamic hand** data for the research of authentication. [9] evaluated their hand authentication technique on a predefined (simple) gesture set, consisting of 15 gestures sourced from 10 participants. [10] used a Kinect sensor to track the users' hands that are following a moving circle on the screen, they collected 20 testers' data for experiments. The datasets mentioned above are of very small scale, and currently not publicly available. Various algorithms cannot be compared uniformly, which hinders the development of gesture authentication research. The Handlogin dataset proposed in [11] is the only public dynamic hand dataset and the most similar to our work. It contains in-air hand-gesture sequences from 21 subjects, each performing 4 different gestures and repeating for 10 times, resulting in total 840 depth sequences. This dataset is the first one which is specially designed for dynamic hand-gesture authentication, but its data size can not satisfy the demand for training data-hungry deep neural networks. Also, because of the limited numbers of testing samples, this dataset can not fully explain the effect and performance of the authentication methods. According to the public information, our proposed SCUT-DHGA dataset is the largest one in dynamic hand-gesture authentication domain. Detailed comparison of our dataset and other related hand-gesture datasets is shown in Table.I.

B. Hand-Gesture Based Authentication Algorithms

Although researches in recent years have mainly focused on applying deep network model [12], [16], [17] and multimodal fusion methods [18]–[20] on classifying hand-gesture, there has been few researches investigating hand-gesture authentication. In total there are mainly two types of approaches for hand-gesture authentication: trajectory-based hand-gesture authentication and vision-based hand-gesture authentication.

Trajectory-based hand-gesture authentication methods utilize movement information of hand keypoints or hand itself to extract features. [9] used a Creative Senz3D sensor to record the 3D locations of users' fingertips, and applied a dynamic time warping (DTW) algorithm to calculate the distances between different trajectories. [10] built different classifiers (Support Vector Machine, Naive Bayes, Classification Tree, K-Nearest Neighbors, Random Forest and Neural Networks) based on 3D position features of hand, which is tracked by a Kinect sensor while following a circle moving on the screen. In [21], Xuan Wang *et al*. used Leap Motion as the input device to continually collect the 3D hand skeleton information, then they used one-class classification with sparse auto-encoder (SAE) to authenticate users. In addition,

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TABLE I

COMPARISON OF CURRENT HAND-GESTURE DATASETS. D AND S INDICATE DYNAMIC AND STATIC DATA TYPE RESPECTIVELY. CLS. AND AUTH. REPRESENT CLASSIFICATION AND AUTHENTICATION RESPECTIVELY. THE DURATION STATISTICS ARE MEASURED USING THE NUMBER OF FRAMES

Dataset	Subjects	Gestures	Repetitions	Total Samples	Modalities			Task	Data Type	Session	Capture Device	Duration Mean/Std	Age Range/Mean	Public
					depth	RGB	others							
DHG-14/28[5]	20	14	5	5,600	✓		skeleton	Cls.	D	1	RealSense	-	-	✓
SKIG[6]	6	10	1	2,160	✓	✓		Cls.	D	1	Kinect v1	145/60.9	-	✓
EgoGesture[7, 8]	50	83	< 1	24,161	✓	✓		Cls.	D	1	RealSense	38/13.9	20~41/25.8	✓
NVGestures[12]	20	25	< 4	1532	✓	✓	Stero-IR	Cls.	D	1 or 2	SoftKinetic (DS325)	80/0.0	-	✓
VIVA[13]	8	19	< 6	885	✓	✓		Cls.	D	1	Kinect v1	31/11.9	-	✓
JUET contact[14]	50	1	10	500		✓		Auth.	S	1	-	-	-	
[15]	4	26	4	416		✓		Auth.	S	1	-	-	-	
[9]	10	15	20	3,000			Trajectory	Auth.	D	1	Creative Senz3D	-	18~46/28.3	
[10]	20	20	1	400			Trajectory	Auth.	D	2	Kinect v1	-	-	
Handlogin[11]	21	4	10	1,680	✓		Optical flow	Auth.	D	1	Kinect v2	-	-	✓
Ours	**193**	**6**	**10**	**23,160**	✓	✓		**Auth.**	**D**	**2**	Kinect v2	64/0.0	19~34/23.0	✓

"signature" type hand-gesture authentication methods have also been proposed using fingertip features extracted by accelerometer, touchscreen or fingertip tracking [22]–[25]. These methods only use multiple or single point coordinate in space and propose elastic matching or classification algorithms for authentication. The physiological shape information of the hand is not used in all these approaches.

Most of the existing **vision-based hand-gesture authentication methods** are developed based on dynamic hand appearances such as silhouette shape and hand geometry. Jonathan Wu *et al*. leveraged a temporal hierarchy of depth-aware silhouette covariance for user authentication [11]. Shefali Sharma *et al*. proposed a multimodal biometric system for personal identity authentication by using hand shape and hand geometry [14]. Some static hand authentication methods based on color histogram, discrete cosine transform, and entropy features also have been investigated [15], [26].

Deep learning methods automatically learning features at multiple levels of abstraction allowing a system to learn complex mapping functions directly from data, without relying completely on human-crafted features. Recent advances in action recognition methods and the introduction of various large video datasets have made it possible to efficiently classify unprocessed streams of visual data with spatio-temporal deep neural network architectures [27]–[31]. Similar to action recognition, deep neural network can also be used to extract features from dynamic-hand-gesture video sequences. Jonathan Wu *et al*. designed a two-stream convolutional neural networks to learn the distinguishing features of different subjects' dynamic-hand-gesture [32]. They passed depth frames one by one to a "spatial stream" convolutional network, and passed optical flow frames one by one to a "temporal stream" convolutional network, and then the outputs of two networks are combined to yield a final output of a video. They used the two-stream CNN for two tasks, namely closed-set identification and open-set authentication. In closed-set identification, they tested performance both when a gesture has been seen by the system and also when it has not. In open-set authentication, they adopted the two-stream CNN model to extract the feature of different types of gestures. They consider all the gestures performed by one subject (no matter what gesture it is), are of the same label. And all the testing gestures have been seen by the model in the training stage.

Different from [32], we conduct experiments on open-set gesture-predefined and gesture-free authentication. These are feature extraction and metric learning problems. It is essentially different from closed-set identification in [32], which is actually a classification problem. In this article, we propose a new benchmark: DHGA-net for SCUT-DHGA dataset, which is the first effort to apply the 3D spatio-temporal deep neural network in dynamic-hand-gesture authentication task.

C. Video Feature Extraction

Generally, there are mainly three video feature extraction methods: 1) To design hand-crafted extraction. 2) Employ 2D ConvNets to extract the features of each video frames and then combine them. 3) Use 3D ConvNets to directly extract the features of video clips. These video extraction methods have been widely researched in action recognition, gesture classification and many other specific domains. Before the deep learning age, many hand-crafted spatio-temporal features [33] and multimodal features [34], [35] have been applied to gesture and behavior recognition. Optical flow is also used to develop video feature extractor [36]. To better take advantage of the time-serials property of video, famous improved dense trajectory algorithm (iDT) has been proposed to describe videos by dense trajectories [37], [38].

At deep learning age, 2D ConvNets which made great success in many image-specific tasks is also employed to extract video features. Karpathy *et al*. used image classification networks to extract features independently from each frame, and then pooled all the predictions across the whole video to get a final video feature [39]. In 2014, a Two-stream ConvNet architecture was proposed, a single RGB frame and a stack of 10 computed optical flow frames were passed to two separate 2D ImageNet pre-trained ConvNet. Their predictions are averaged to get a final feature of a video [40]. Bilen *et al*.

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Fig. 2. A sample of dynamic hand-gesture collection. (Left) Subject performs the gesture in front of the sensor. (Right-top) The first person perspective. (Right-middle) The Kinect V2. (Right-bottom) The RGB image and its corresponding depth image during data acquisition.



Fig. 3. The 6 classes of hand-gesture designed in our proposed SCUT-DHGA dataset. Each row shows a sequence of selected key frames of a dynamic-hand-gesture. Yellow arrows indicate the finger motion of that frame.

improved the feature extraction performance of two-stream network in terms of new representations [41], [42].

A more direct idea to extract video features is to leverage 3D convolutional networks. Some work applied 3D convolutional networks to directly create hierarchical representations of spatio-temporal data [27], [43], [44]. I3D [28] "inflate" the weights of 2D convolutional filters in an Inception V1 architecture [45] into 3D convolutions. It initializes the inflated model weights by duplicating weights that are pre-trained on ImageNet classification over the temporal dimension, therefore to get a better feature extraction result.

Most recent work on improving video feature extractor is focusing on extracting video feature in different frame rates [46], involving more frames in video [47].

To develop a dynamic-hand-gesture authentication method, it is of highly importance to design a "good" feature extractor.

III. The SCUT-DHGA Dataset

A. Data Collection

When collecting dynamic-hand-gesture data, as shown in the left part of Fig.2, volunteers were asked to perform the gestures in front of the sensor within the distance of 0.6-0.8m to ensure the integrity of depth images. Before collecting each type of gesture video (total six types), volunteers were told a short description of that gesture and shown a template video. After they learned and got familiar with that gesture, they performed it for 10 times. Between each time of collection, volunteers were required to move their hand out of the recording area and then move in again, to provide data with more diversities. Also, between collecting each type of gesture video, volunteers were asked to relax their hand to keep in good condition in order to avoid performing unnatural gesture. With an average interval of one week, some volunteers showed up for the second session data collection.

B. Dataset Characteristics

Subjects: The research of hand-gesture authentication requires enough subjects to perform same hand-gesture rather than few subjects to perform various kinds of hand-gesture. Therefore, we invited 193 volunteers for the dataset collection, including 64 females and 129 males. The average age of all volunteers is 23, the minimum age is 19, and the maximum age is 34. Currently, our SCUT-DHGA dataset is the largest dataset in dynamic-hand-gesture authentication domain.

Gesture Type: In gesture classification task, gestures are mainly used for communication and manipulation, the major principle of gesture design is that gestures should be meaningful and simple. These gestures (for example, open/close, move, push) commonly only involve a slight rotation of palm and simultaneous movement of fingers. In this work, gesture authentication task focuses on behavioral and physiological characteristics of gestures. Since the gestures are used for identity differentiation, they should involve enough fingers' movement of the fingers which contain personal traits. Besides, they also should be user friendly for reproducibility. Under these principles, we conducted a user study with participants to investigate their preferred hand gestures from a candidate set that was collected from various dynamic hand gesture datasets before collecting the dataset. The six top-rated hand gestures shown in Fig.3 were finally chosen for evaluation. They are: **1)** make a fist (start from thumb), **2)** rotate hand while making a fist (start from little finger), **3)** catch and then release, **4)** four fingers (forefinger, middle finger, ring finger, little finger) touching thumb one by one, **5)** bend four fingers(little finger, ring finger, middle finger, forefinger) one by one, **6)** open a fist (from thumb). According to participants' feedbacks, their most comfortable average duration for performing these hand gestures is about 4s. So, for convenience, we set our program to automatically record each gesture for about 4s once it detects the gesture starts.

Session: Time span is very likely to introduce variations into ones' dynamic-hand-gesture, which can be extremely harmful to a dynamic-hand-gesture authentication system. The variations may caused by uncertain position of the hand, illumination changes and so on. To analyze this problem and to better evaluate a dynamic-hand-gesture authentication algorithm, the second session data of SCUT-DHGA dataset is acquired. Totally, 50 of the volunteers' two session hand-gesture data were recorded with about an average interval of one week. When recording the second session data, volunteers were shown template videos as same as the first session did, and then performed the gestures.

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Data Format and Details: To collect the dataset, we select Kinect v2 as our camera because it integrates both RGB and depth sensors. The two modality videos are recorded simultaneously with the full resolution of Kinect v2 and a 200*200 ROI of hand region is cropped for each frame. The RGB image sample and depth image sample are shown in the right column of Fig.2. Note that most of the 3D ConvNets use the videos with 64 frames as inputs for a good trade-off between efficiency and accuracy, we also follow this empirical principle for recording our dataset. As mentioned before, the participants' most comfortable average duration for gesturing is about 4s. So, we use a frame of 15fps to capture for about 4s obtaining videos with nearly 64 frames such that we don't need to trim the videos carefully, which is more efficient and robust. All RGB images and their corresponding depth images have been aligned after collection. As for second session data, a total of 50 volunteers turned up for the videos, so the second session data contain 6,000 video sequences including RGB and depth modality. Totally 29,160 video sequences have been acquired from 193 volunteers for both RGB and depth modalities.

C. Evaluation Protocol

Metric: We use EER (error equal rate) as the evaluation metric. EER is a balance of FAR (The false acceptance rate) and FRR (false reject rate), which is a widely used metric in biometrics. FAR is the probability that the biometric security system will incorrectly accept an access attempt by an unauthorized user. A system's FAR is typically stated as the ratio of the number of false acceptances divided by the number of identification attempts. Meanwhile, the FRR is the probability that the biometric security system will incorrectly reject an access attempt by an authorized user. A system's FRR is typically stated as the ratio of the number of false recognition divided by the number of authentication attempts. Usually, these two rates will have performance trade-offs which can be obtained by applying various matching thresholds. The EER is the error rate when the FRR equals the FAR and adopted to evaluate the matching accuracy with different parameters, because the FRR and FAR are treated equally. An algorithm with a lower EER can exhibit better performance in authentication tasks. Since it is the most common metric for user authentication, we adopt it in the following experiments.

Experiment Setup: We design experiments to set a baseline of two kinds of **open-set** authentication scenarios: gesture-predefined authentication and gesture-free authentication. Gesture-predefined authentication means the gesture type used in the testing stage is the same as that in the training set. We consider the most difficult situation that all the users apply the same gesture type in the testing stage. Gesture-free authentication means all users can use their self-defined hand-gestures as their passwords in the testing stage, which is more user-friendly and close to real-world applications, thus hits more challenging. We consider the most difficult situation in gesture-free authentication that all the users apply the same gesture type, and the testing gesture type should not appear in the training set. As one kind of behavioral biometrics identifiers, the dynamic hand gesture is flexible and suffers from the problem of variation over time. So, we also conduct cross-session experiments where the template gestures and the query gestures are from different periods (sessions) to investigate the effects of this factor on the hand gesture authentication system.

Note that open-set protocol where training users and testing users are not overlapping is more challenging than its close-set counterpart while being more applicable to real applications. The reason is that it is non-trivial to retrain the feature extractor when new users join in. The key is to design effective feature extractor for robust and discriminative gesture feature extraction. Also, it is the most challenging but representative setting when all subjects use the same gesture type for authentication since it requires the feature extractor to focus on the user-specific behavioral characteristics rather than the differences between different gestures.

IV. Benchmarking the Proposed SCUT-DHGA Dataset

In the dynamic-hand-gesture authentication community, there isn't a well recognized deep learning method, so we propose our **D**ynamic-**H**and-**G**esture **A**uthentication network (**DHGA**-net) as a baseline method. In this section, the architecture and the implementation details of **DHGA**-net will be discussed.

A. Baseline Method

As depicted in Fig.4, the DHGA-net architecture consists of two major components: a spatio-temporal feature extraction backbone, and a **T**emporal-**I**dentity-**E**xtrating module (**TIE**). Unlike some temporal-segmentation based methods used for action recognition and gesture recognition, which sparsely sample video frames to form the network input, dynamic-hand-gesture authentication should take densely sampled frames as input to gather more dynamic information. A 3D convolution network can take the densely sampled video frames as input and extract the spatial and temporal features that corresponds to the physiological and behavioral traits simultaneously. We design a spatio-temporal feature extraction backbone based on 3D convolution layers to extract spatio-temporal features of dynamic hand gesture videos. We also design a TIE module to handle the gesture features extracted from the spatio-temporal feature extraction backbone.

1) Spatio-Temporal Feature Extraction Backbone: is modified from I3D [28], which is used for the task of action classification. I3D maps a video to a single feature vector and classifies it directly. Dynamic-hand-gesture authentication is a metric learning task that is different from classification task. To extract identification feature with more discriminative information, we modify the origin I3D network by discarding some deep layers to reduce the number of parameters as well as shrink the reception along the time axis. It consists of three 3D convolutional layers, four 3D max-pooling layers, and six 3D inception blocks. As shown in Fig.4, a t frames RGB or depth hand-gesture video with a shape of $(t \times 200 \times 200 \times 3)$ in SCUT-DHGA dataset is fed into the feature extractor, whose

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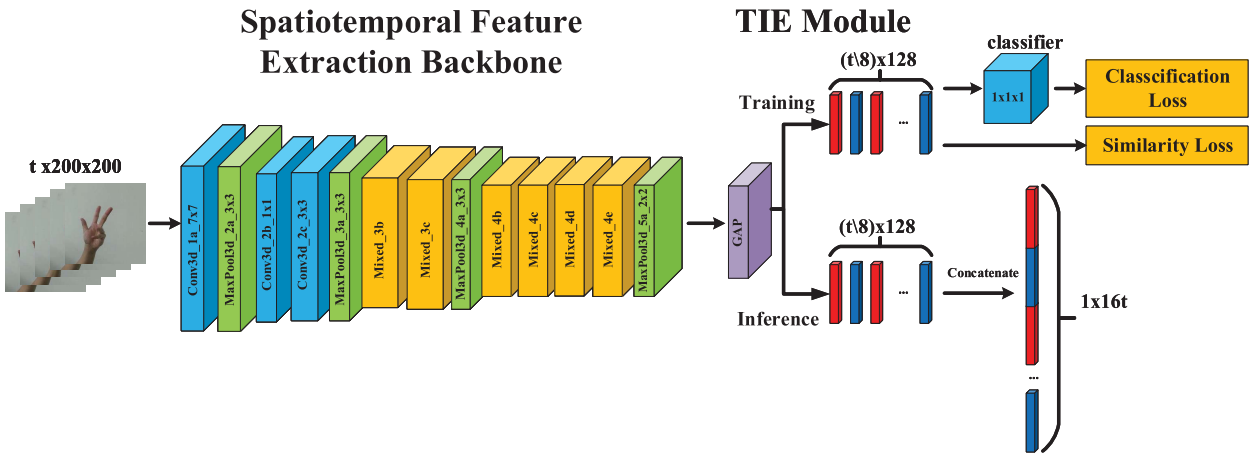


Fig. 4. The DHGA-net consists of two parts: spatio-temporal feature extraction backbone and TIE module.

output is a feature with a shape of $(\frac{t}{8} \times 7 \times 7 \times channels)$. And then, this feature will be fed to the TIE module.

2) TIE Module: is built to take fully advantage of video data for authentication task. Different from hand-gesture classification system which need to "see" all the frames of a video to recognize a hand-gesture, a dynamic-hand-gesture authentication model intuitively can authenticate a user just by observing a segment of a video, because even a small segment of a video contains rich information of physiological and behavioral characteristics. The TIE module is shown in Fig.4, it behaves differently in the training and testing stage.

In the training stage, the feature extracted by the spatio-temporal feature extraction backbone is sent to a Global Average Pooling layer [48] to integrate spatio information. The output feature's shape of the GAP layer is $(\frac{t}{8} \times channels)$, we treat it as a feature sequence, the length of this feature sequence is $(\frac{t}{8})$ and size of each feature is $(channels)$. Here, each feature in the sequence encode the same user's identity trait, but each of them can only observe a segment of the video along the time axis because of the reception field constrain of the convolution operation. In other words, each feature represents different parts of a same dynamic-hand-gesture video but can enough represent the identity information of this video. The advantage of TIE module is that it plays a role as data augmentation in deep feature level.

A modified joint supervision of softmax cross-entropy loss and center loss [49] is applied to supervise the net to obtain better intra-class compactness as much as possible. The loss function is shown in Eq.1.

$$\mathcal{L} = -\sum_{a=1}^{k}\left(\sum_{j=1}^{t}\log\frac{e^{W_{y_{ij}}^{T}x_{ij}+b_{y_{ij}}}}{\sum_{i=1}^{n}e^{W_{y_{ij}}^{T}x_{ij}+b_{ij}}} + \lambda\sum_{i=1}^{t}\left\|x_{ij}-c_{y_i}\right\|_2^2\right) \tag{1}$$

$x_{ij} \in \mathbb{R}^d$ denotes the ith sample in a mini-batch and jth feature in the feature sequence. The $c_{y_{ij}} \in \mathbb{R}^d$ denotes the class center of deep features of x_{ij}. W denotes the weights of the last 3D convolution layer which plays a role as a classifier, and b is the bias. λ denotes the balancing factor between classification loss and similarity loss, we set $\lambda = 0.001$ in all our experiments.

In the testing stage, DHGA-net plays a role as a feature extractor. After the GAP layer, features in the feature sequence are concatenated to form a final feature. We calculate the cosine distance between two features (query gesture and registered gesture) to represent their similarity. If two features have a smaller distance than a threshold, which means they are similar, they should come from the same subject, otherwise they come from different subjects.

B. Implementation Details

Although our SCUT-DHGA dataset is the largest one in dynamic-hand-gesture authentication domain, there are still not enough training samples to train all the weights of a deep 3D convolution network from scratch. Therefore, we use ImageNet pretrained weights for model initialization, then fine-tune the DHGA-net with our dataset. We train the DHGA-net for 30 epochs with a batch size of 8 on a single GTX1080Ti GPU. We set the initial learning rate to 0.1 and decrease this rate by one-tenth at the 15th epoch. We use SGD as optimizer, the momentem and weight decacy are set to 0.9 and 1e-7 respectively. For data augmentation, we randomly apply gamma correction($\pm$30%), color jittering($\pm$15) and normalization on each training dynamic-hand-gesture video. Our DHGA-net is built using Pytorch framework [50], [51]. The code will be released soon. ID in the range of (0-142) are split into the training set and (143-192) are split into the testing set. The testing set consists of two sessions' data from two different periods of time.

V. Experiments and Analysis

As aforementioned in section III, RGB and depth modalities data are collected in SCUT-DHGA dataset, we use two kinds of data format for experiments: **dynamic RGB data** (RGB videos), and **dynamic depth data** (depth videos). In this section, we conduct three major experiments to investigate the performance and challenge of dynamic-hand-gesture authentication. The first and second experiments adopt dynamic RGB

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TABLE II
SETUPS FOR SINGLE-SESSION AUTHENTICATION

Experiments ($i \in [1-6]$)	Training Gestures	Test Gesture
$SG_gesture_i$	$gesture_i$	$gesture_i$
$MG_gesture_i$	All Gestures	$gesture_i$
$USG_gesture_i$	All(separately) except $gesture_i$	$gesture_i$
$UMG_gesture_i$	All(together) except $gesture_i$	$gesture_i$

data as training and testing data: **1)** The first experiment aims to test the basic performance of dynamic-hand-gesture authentication. **2)** The second experiment is designed to investigate whether the time-span will influence authentication performance, and to discuss the plausible reason and its internal mechanism. It is of vital importance to conduct this experiment because cross-session authentication simulate the real world authentication situation. **3)** The third experiment adopts dynamic RGB and depth data aiming to study the advance of using multi-modalities fusing strategies to improve the authentication performance.

A. Single-Session Authentication

The main objective for this experiment is to test the basic performance of hand-gesture-authentication, and to ascertain the feasibility of gesture-predefined authentication and gesture-free authentication. Gesture-predefined authentication means the gesture used for testing has been seen by the model at training stage. Gesture-free authentication means the gestures used for testing are different from gestures used for training.

The setups of single-session experiments are shown in Table.II. SG indicates "single gesture", means that model is trained using only one gesture, and tested on the same gesture. For example, in $SG_gesture_1$ setup, we only use the $gesture_1$ data for training and testing.

Experiment setup MG indicates "multiple gestures", we train a model using all the six gestures together and test the authentication performance on six gestures separately. The main difference between SG and MG is that MG setup trains the model with multiple gestures, while SG trains model with less gestures. These two experiments' setups mentioned above belong to gesture-predefined authentication, because the testing gesture has been seen by the model.

Experiment setups USG and UMG indicate "unseen single gesture" and "unseen multiple gestures", they belong to the gesture-free authentication. The testing gestures are strictly disjointed from training set, therefore to investigate the generalization performance of dynamic-hand-gesture on unseen gesture types.

In USG setup, we train five models separately with five different gestures and test them on the left one by averaging their results. For example, in $USG_gesture_1$, five networks are trained separately on five different gestures ($gesture_2$ to $gesture_6$), then we test these five models with $gesture_1$ separately and average the result to get the final EER.

As for UMG setup, we train the model on five of the six gestures together, and test the performance on the left gesture. For example, in $UMG_gesture_1$, the network has been trained on multiple gestures expect $gesture_1$, then test on $gesture_1$.

The results of these four experiments are shown in Fig.5. From the experimental results, no matter on single gesture or their average performance, it suggests that:

1) dynamic-Hand-Gesture (Even Everyone Use a Same Gesture) is a Discriminative Biometric to Distinguish Identities: The average EER of four experiments are: MG(0.97%), SG(2.14%), UMG(2.27%), and USG(3.20%), which indicate that, after training, a model has the ability to extract user's unique discriminative feature from their dynamic-hand-gesture even in different scenarios and experiment setups.

2) Involving More Gesture Types in Training Set can Effectively Improve the Performance of Dynamic-Hand-Gesture Authentication: The difference between experiment setup MG(0.97%) and SG(2.14%) is that MG involves six dynamic-hand-gesture type but SG only involves one. Also, UMG involves more gesture types than USG. It is clearly that the result of MG is better than SG, and the result of UMG is better than USG. One possible explanation is that by training the DHGA-net using data with more gesture types, the model is able to extract more discriminative features from dynamic-hand-gesture.

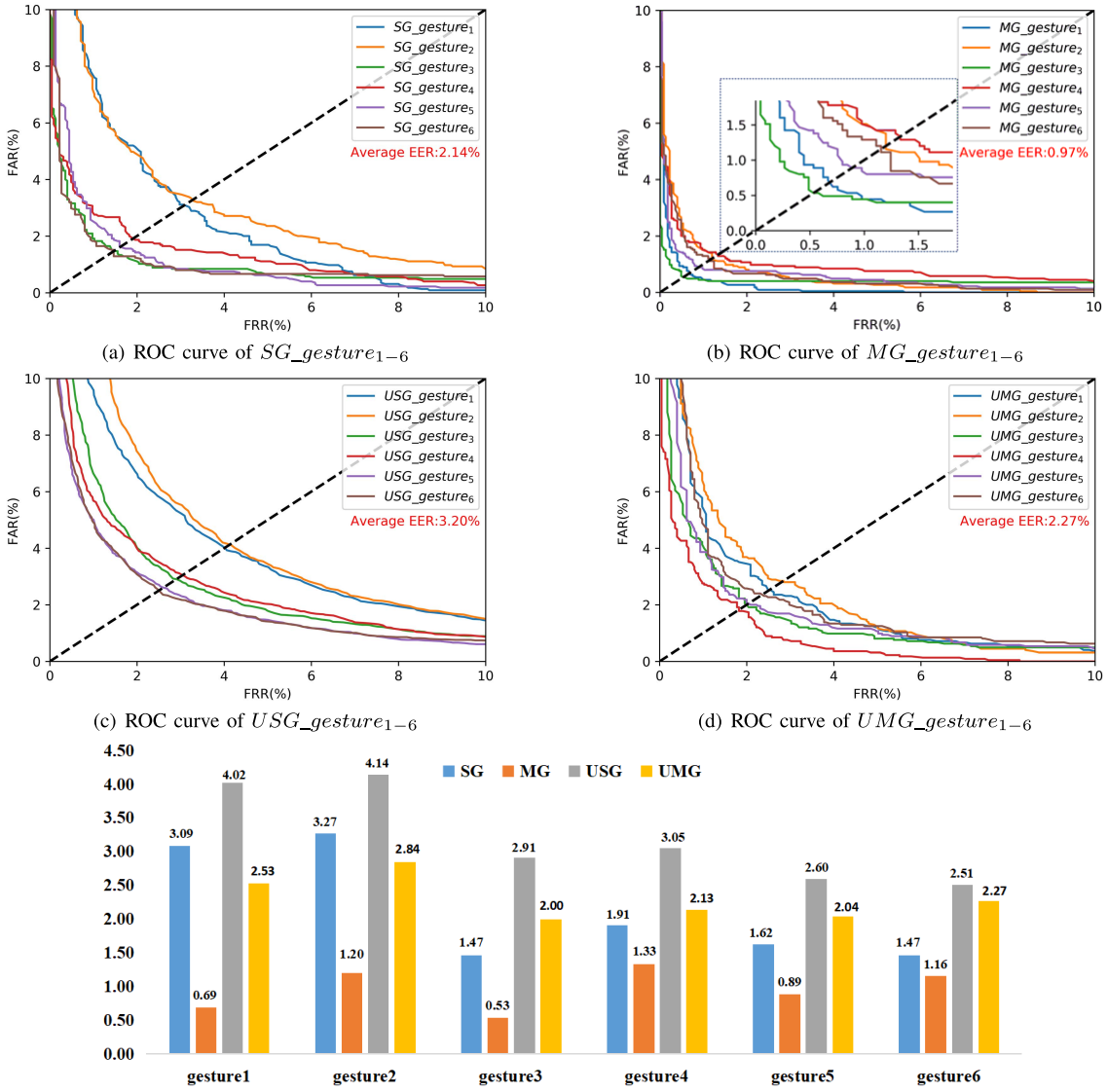
3) Difficult Dynamic-Hand-Gestures Have Better Authentication Performance Than Easy Ones: As shown in Fig.5, compared with other gestures, $gesture_1$ and $gesture_2$ have poorer performance. These two hand-gestures are more frequently used by people and easier to perform than other gestures. Such gestures done by different subjects contain similar trajectories and dynamic information, which may cause to less distinguish behavioral features. Comparatively, $gesture_{3-6}$ have better authentication performance because they are more difficult, therefore subjects are more likely to show distinguish physiological traits(from different frames) and motion traits (across contiguous frames) when performing gestures.

4) Gesture-Free Authentication is More Practical Yet More Challenging: By comparing SG(2.14%) to USG(3.20%), and MG(0.97%) to UMG(2.27%), it is easily to tell that a model which tested on an unseen gesture has a worse performance than a model tested on a gesture that has been seen in training stage. After training, the model is good at extracting the identity features from the specific gestures that have been trained on. Nevertheless, it still gains the ability to extract identity features from unseen gestures, which indicates that the model has the ability to extract the uniform physiological characteristics and behavioral characteristics from different unseen gestures.

B. Gesture Variation Over Time

The main objective of this experiment is to investigate the authentication performance when query gesture and registered gesture come from different periods of time. This task is a very challenging yet having practical significant. It is obvious that time-span will affect the repetition of performing the same gesture. In most behavior-based authentication, such as signature, gait and so on, the time-span is a very important issue [52]–[54]. Especially, when the time-span is

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(a) ROC curve of $SG_gesture_{1-6}$

(b) ROC curve of $MG_gesture_{1-6}$

(c) ROC curve of $USG_gesture_{1-6}$

(d) ROC curve of $UMG_gesture_{1-6}$

(e) experiment result of different gestures in SG, MG, USG,experiment

Fig. 5. The ROC curve and statistic histogram of single-session experiments.

large, behavior characteristics may be very different. Therefore, it is essential to evaluate a behavior-based authentication method on data from multiple sessions. This experiment also contributes to follow-up work to better investigate the gesture variation over time.

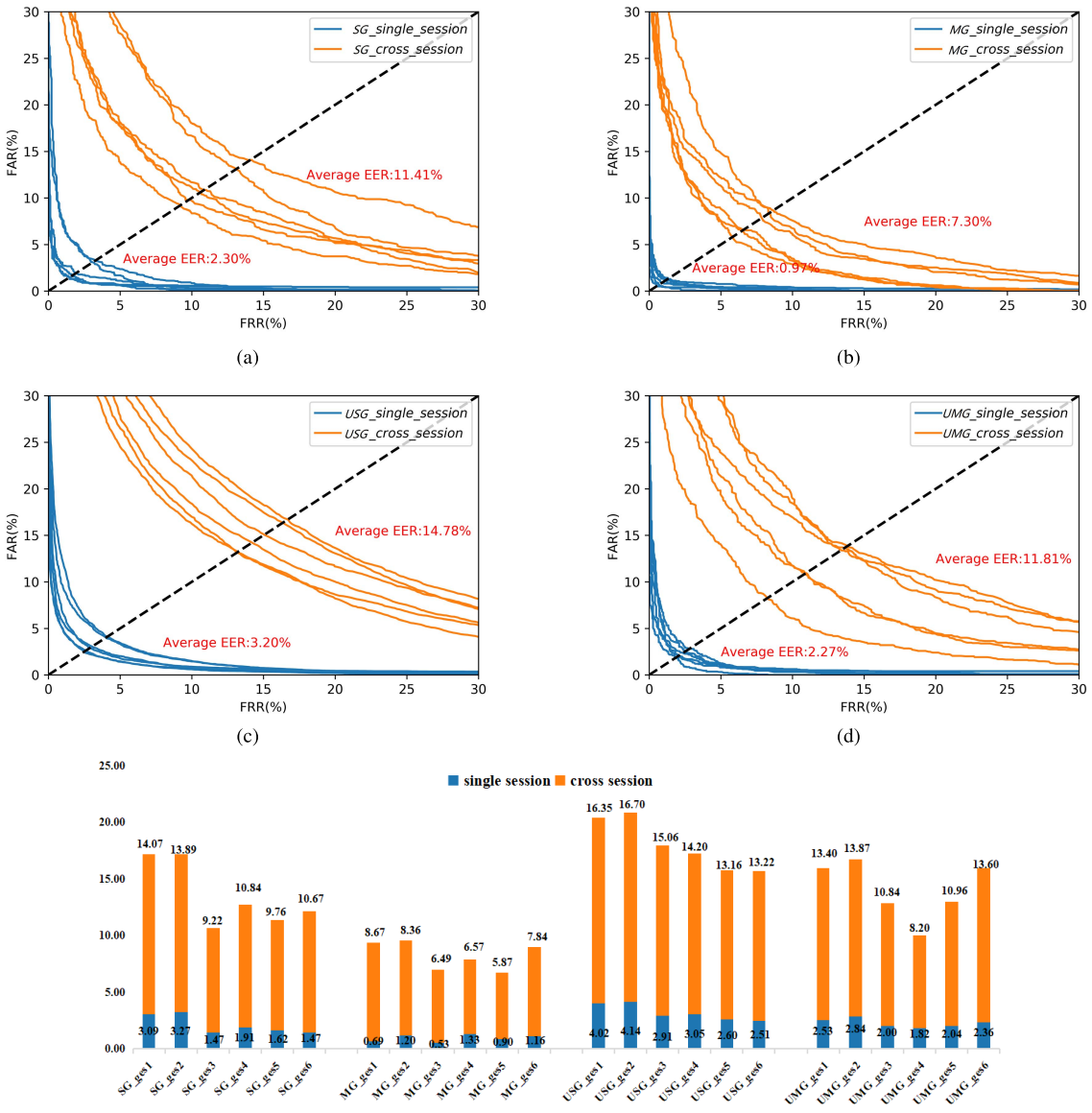
The average time interval of query and registered samples we used in SCUT-DHGA is one week. We conduct this cross-session experiment under SG, MG, USG and UMG setups. The experiment results are summarized in Fig.6.

Unsurprisingly, there is an obvious performance degradation when authenticating on cross-session dynamic-hand-gesture data. As shown in Fig.6, the average performance drop of using cross-session data are: 9.11% under SG setup, 6.33% under MG setup, 11.58% under USG setup, and 9.54% under UMG setup. All the results prove that authenticating cross-session data is much more challenging than authenticating single-session data. There are two plausible factors to explain this phenomenon: environmental variation and gesture variation.

1) Environmental Variation: When authenticating dynamic-hand-gestures collected at different periods, environmental changes often cause performance degradation. Time span mainly incurs illumination variation, distance and angle variation in the collection progress, further leads to the variation of the data. A typical environmental change: illumination variation, is well recognized to influence the feature extraction.

2) Gesture Variation: Different from repeatedly entering a password, when a person performs gestures, there must be subtle differences (e.g. speed and angle of gestures) in each repetition. Firstly, it is obvious that no one can make an exact same gesture for a second time, it must be a slight difference between two dynamic-hand-gestures even they are performed by the same subject. Secondly, there exists position and angle variations in each recording, because we didn't make position and angle restrictions when subjects were performing dynamic-hand-gesture. Such differences are affected by many factors, such as specific emotional state (for example, stress, anger or sadness, etc.), physical state

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(e) Experiment result of different gestures in SG, MG, USG, and UMG experiment

Fig. 6. The ROC curve and statistic histogram of cross-session experiments.

(for example, illness, tiredness, etc.), and forgetting caused by time-span.

Both factors may have a great impact on the distribution of the captured hand gesture videos, so does the distribution of the extracted features from the backbone network, and then the performance of gesture authentication may degrade across different periods of time. To demonstrate this speculation more clearly, we visualize the feature distributions of 50 users' hand gestures from two different periods of time using t-SNE [55]. The result is shown in Figure.7. It can be seen that the feature distributions indeed variate a lot over time, which makes the authentication over time, that is, the templates and queries are from different periods of time, more difficult. For example, for a pair of subjects #17 and #46, their features are highly separable for a single period of time, while some samples of #17 in the second session are closer to some samples of #46 in the first session, when those samples are selected queries and templates the authentication failed. Template update may be a good strategy to alleviate this problem since the feature distributions of gestures from a single period of time are compact. We test a basic template update strategy to demonstrate its effectiveness. Specifically, we first train the proposed DHGA-net using the data collected from the first session and then use this trained model to extract features for gestures captured in the second session, the templates and queries are both from the second session. This basic method can achieve 0.64% average EER under the *MG* setting. More sophisticated template update strategies and other alternative solutions, including generative models, can be explored to improve the performance of hand gesture authentication over time, which is out of the scope of this work.

Apparently, better quality data that are collected with more constraints would alleviate the impact of environmental and gesture variation but will reduce the usability of a gesture authentication system. There exists a trade-off between

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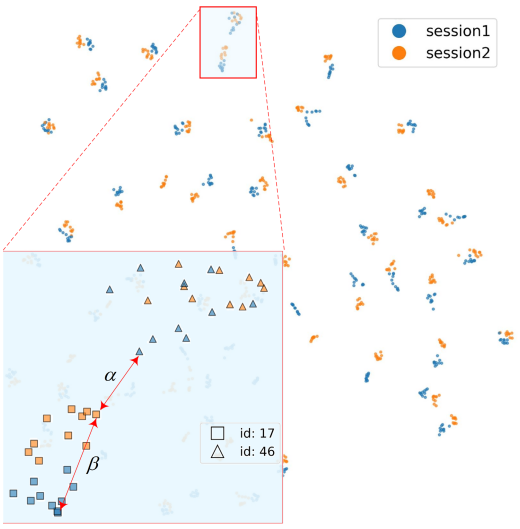


Fig. 7. Feature distributions of hand gestures from two different periods of time.

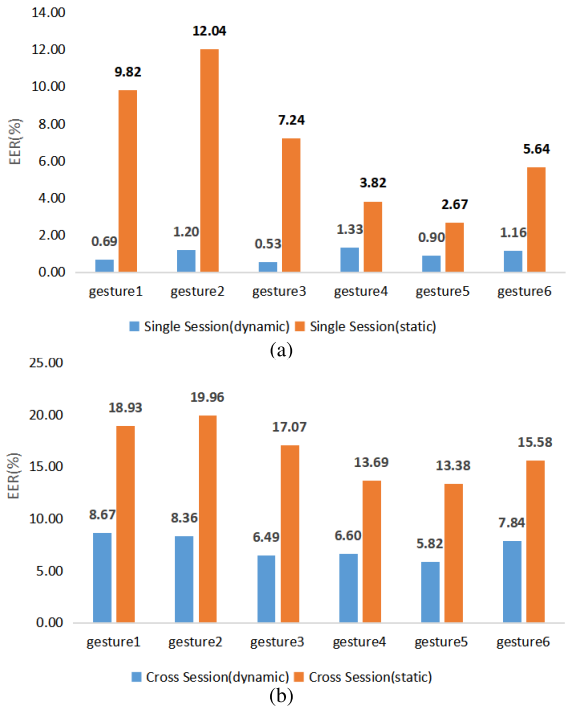


Fig. 8. Comparison of authenticating using static data and dynamic data.

usability and security. So it is of vital importance to evaluate an authentication method using cross-session data.

C. Importance of Using Dynamics Information

To demonstrate the importance of dynamic characteristics of gestures for authentication, we conduct a contrastive experiment that only uses the static hand gestures to authenticate identity. For fair comparison, we use the same DHGA-net as the feature extractor and keep other experimental settings unchanged to test the authentication performance of static-hand-gestures on the SCUT-DHGA dataset. The static samples are created by copying and stacking single frames of the dynamic-hand-gesture videos to 3D tensors that have the same shape as original video inputs to the DHGA-net. The results

TABLE III
COMPARISON WITH TS-CNNS ON SCUT-DHGA

		Equal Error Rate(%)						
Gesture		g1	g2	g3	g4	g5	g6	Average
Single Session	TS-CNNs	13.69	10.93	12.84	12.89	11.29	10.36	12.00
	DHGA-net	**1.55**	**3.11**	**1.78**	**1.56**	**1.96**	**1.51**	**1.91**
Cross Session	TS-CNNs	31.60	29.96	32.13	32.24	33.42	33.60	32.16
	DHGA-net	**13.82**	**13.49**	**12.04**	**12.53**	**13.64**	**13.31**	**13.14**

TABLE IV
COMPARISON WITH TS-CNNS ON HANDLOGIN

	Equal Error Rate(%)				
Gesture	gesture1	gesture2	gesture3	gesture4	Average
TS-CNNs	0.44	1.29	4.89	1.05	1.92
DHGA-net	**0.00**	**1.67**	**1.40**	**0.00**	**0.77**

in Fig.8 show that the dynamic gestures outperform their static counterparts significantly for both single-session and cross-session tests, which demonstrates that the dynamic characteristics of gestures have a significant advantage for authentication.

D. Comparison With the State-of-the-Art

As far as we know, the two-stream CNNs (TS-CNNs) [32] is the only existing deep learning-based method for gesture-based authentication, which achieves state-of-the-art performance over previous conventional methods on a small hand gesture dataset Handlogin [11]. So, we compare with the TS-CNNs to provide more benchmark tests and demonstrate the superiority of our proposed DHGA-net. The TS-CNNs uses a spatial stream 2D ConvNet and a temporal stream 2D ConvNet to extract spatial and temporal feature separately, the inputs to those two networks are original video frames and their corresponding optical flow maps respectively. The extracted features from two ConvNets are concatenated for authentication. The comparisons are conducted on both Handlogin and our SCUT-DHGA datasets. The results are reported in Table III and IV.It shows that our DHGA-net outperforms TS-CNNs by a large margin, especially on our SCUT-DHGA dataset (1.91% *vs.* 12.00% average EER for single session test, and 13.14% *vs.* 32.16% average EER for cross session test). A reasonable explanation may be that the way of separated spatial and temporal feature extraction from single frames cannot able to capture effective features representing user identity. While our DHGA-net can capture more effective spatial and temporal features with a 3D ConvNet backbone since the 3D convolution is capable of fusing multiple frames' features adaptively.

E. Comparison of Different Multi-Modal Fusion Strategies

This set of experiments is for comparing authentication performance between different multi-modal fusion strategies. For multi-modal fusion, the model obtains the multi-modal data and learns to fuse them at different level, including:

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TABLE V
RESULTS OF MULTI-MODEL FUSION EXPERIMENTS

	Gesture	Equal Error Rate(%)							
		Feature Level Fusion (RGB,Depth)						Data Level Fusion	Score Level Fusion
		(1,0)	(0.8:0.2)	(0.6:0.4)	(0.4:0.6)	(0.2:0.8)	(0:1)		
Single Session	gesture1	0.69	0.58	**0.40**	0.58	1.33	1.55	2.04	**0.36**
	gesture2	1.20	1.24	**1.20**	1.96	2.93	3.11	2.00	1.53
	gesture3	**0.53**	0.58	0.58	0.84	1.47	1.78	2.31	0.67
	gesture4	1.33	1.24	1.02	**0.76**	1.33	1.56	1.24	0.89
	gesture5	0.89	0.84	**0.76**	0.87	1.64	1.96	1.33	**0.67**
	gesture6	1.16	0.98	**0.67**	0.84	1.16	1.51	1.42	0.73
	Average	0.97	0.91	**0.77**	0.98	1.64	1.91	1.72	0.81
Cross Session	gesture1	8.67	8.67	**8.76**	10.22	13.07	13.82	11.33	9.22
	gesture2	8.36	8.36	**7.78**	9.67	12.93	13.49	12.58	8.31
	gesture3	6.49	6.31	**5.58**	7.78	10.98	12.04	9.47	6.02
	gesture4	**6.60**	6.60	6.33	8.84	11.78	12.53	11.20	7.36
	gesture5	5.82	5.73	**6.13**	8.49	12.53	13.64	9.24	6.67
	gesture6	7.84	7.69	**7.20**	9.22	12.36	13.31	10.04	7.69
	Average	7.30	7.23	**6.96**	9.04	12.28	13.14	10.64	7.55

(1) early fusion (i.e. data level); (2) intermediately fusion (i.e. feature level); (3) late fusion (i.e. score level).

Generally, the RGB data contain more identity information such as palm-print and hand texture. However, depth data may provide extra information which is helpful for authenticating. First, we experiment on RGB and depth modality data separately, and then test different multi-model fusion strategies data level fusion, feature level fusion and score level fusion.

In data level fusion experiment, RGB video and depth video are stacked together and then directly sent to DHGA-net. In feature level fusion experiment, RGB video features and depth video features are extracted separately by two different DHGA-net, these two features are concatenated to form the final features. In score level fusion experiment, the final distance between register sample and query sample is the sum of the distances of RGB modality and depth modality. Experiments are conducted on single-session and cross-session authentication under MG setup, to test the best performance of dynamic-hand-gesture authentication. The result is shown in Table.V

Table.V shows the authentication EER for different modalities and multi-modal fusion strategies. The column (1:0) shows the performance of only using RGB modality feature. The column (0:1) shows the performance of only using depth modality feature. The columns between (1:0) and (0:1) show the performance of feature level fusion with different balance factor. The second column from the right is the results of score level fusion. And the last column is the results of data level fusion.

Across all experimental results, multi-modal fusion at feature layer with a (0.6:0.4) weighted concatenating has the best performance. Score level fusion perform well in some gestures, but totally, feature level fusion is still more stable and usable. Lastly, Data level fusion performs just slightly better than only using depth modality data.

One potential explanation is that, depth data contain additional three-dimensional spatial information, which is complementary to RGB data. An appropriate fusion of depth features can improve the performance of authentication (as the column for (0.6,0.4) shows). Moreover, if only using depth modality data for authentication, or involving excessive depth feature, the performance gets worse. This may be attributed to the fact that the depth information itself contains relatively less identity information, such as palm texture. Besides, the noise in the depth data, which caused by the sensor during the acquisition process is also one reason for the poor performance. We also notice that the multi-modal fusion at data layer yields no better performance than single RGB modality. Because the original data of early fusion usually contains a lot of redundant information that make model harder to train.

VI. CONCLUSION

In this article, we proposed SCUT-DHGA, a new dataset for dynamic-hand-gesture authentication task. The proposed SCUT-DHGA contains 193 subjects and total 29,160 video sequences including both RGB modality and depth modality data. Especially, it includes two sessions dynamic-hand-gesture data to investigate the gesture variation problem caused by time span. Besides, we clearly define the evaluation metric for dynamic-hand-gesture authentication task. Also, we have proposed a novel DHGA-net, to extract discriminative information from dynamic-hand-gesture video for authentication, and benchmark the SCUT-DHGA dataset with our proposed method. The best EER result we can achieve now is 0.77% in single-session scenario and and 6.96% cross-session scenario, by fusing the RGB and depth data in feature level. Obviously there is still substantial room to improve the performance on this dataset which can also support various potential research directions.

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